ActInf Livestream #018.1 ~ The predictive global neuronal workspace: A fo

Livestream | The predictive global neuronal workspace: A formal act inf model | 1:57:41

hello and welcome
to the active inference lab this is the
active inference lab and the active
inference live stream
today we are in active inference live
stream 18.1
on march 23rd 2021
welcome to the active inference lab
everyone we're a participatory online
lab
that is communicating learning and
practicing applied active inference
you can find us at our links here this
is a recorded in an archived live stream
so please provide us with feedback so we
can improve our work
all backgrounds and perspectives are
welcome here and we'll be
using good video etiquette for live
streams
today we're in live stream 18.1 in our
first of two participatory group
discussions
on the predictive global neuronal
workspace paper and we're with the
authors here
ryan smith and christopher white so
thanks to both of you
for joining today today in 18.1
the goal is just going to be to discuss
and learn and ask some questions about
this awesome paper and we're going to go
through some introductions and warm-ups
and then
i believe christopher will have some

slides to share with us
and set some context and then we'll just
proceed with
probably a bunch of questions that we
all have about the paper as well as some
questions from the live chat
to get started though let's just go
around
and introduce ourselves and we can each
just give a short introduction or check
in and then pass it to somebody who
hasn't spoken
so i'm daniel i'm in california and i
will pass it first to dean
i'm dean i'm up here in calgary in
canada
and i'll pass to stephen
hello i'm stephen i'm up in toronto in
canada
and i'll pass it over to christopher
hi thanks i'm christopher white i'm a
phd student at
the university of cambridge and this is
this page just to give some brief
context this paper is really
a work that kind of came out of my
master's thesis that i was doing with
ryan but we've kind of continued on
um working on so hopefully we can chat
about both this paper and then at some
point
chat about some things that we've been
continuing on
and so i'll talk past to blue
hi i'm blue knight i am an independent
research consultant based out of new
mexico
and i will pass it to dave

dave douglas i'm in the mountains of the
northern philippines where it's
almost the end of spring um
i'm retired from i.t uh especially from
uh machine
translation working on the
semi-technical aspects of our education
project
and then i believe ryan um
yeah so i'm ryan smith um i'm an
investigator at the
laureate institute for brain research um
and i
yeah we're in tulsa oklahoma basically i
run a lab focused on
computational neuroscience and
computational psychiatry
well i'm sure we'll talk a lot more
about that let's uh
maybe just go right to christopher's
slides
and then we can pick up just with
conversation or with this
slide deck as needed but christopher go
ahead and share your
slides and i'll make them the full
screen yeah sure
okay
okay so i'm assuming you can see my um
screen are you saying we're seeing our
jitsi
uh okay how about now
yep looks good okay well i now i it
seems like i can't
both look at the even of the dual
monitor setup it doesn't seem like it
it wants to share the same screen that
jitsi is on and that's okay so it means

i just can't see anyone
um so just give me a shout
um so this this is these are slides that
uh from a talk i gave at ucl
about midway through last year um
and so i i may need some reminding of
bits of this paper because we did write
it quite a while ago although it did get
only just get published in december last
year
okay so the idea with this paper was
really just to
kind of formalize some things
that had been around in literature for a
while
and namely putting global workspace
theory
into kind of a predictive processing
framework
and so they've been so carfriston has a
really nice 2012 paper that kind of did
this
somewhat formally uh jacob but it didn't
really connect that deeply with the
global workspace literature
um at all really um jacob had
in chapter 10 of his 2013 book yakupovi
sorry
um has a really wonderful chapter where
he discusses
the relationship between phenomenal
unity and global workspace theory and
how
active and how predictive kind of the
predictive processing framework in
general and active inference
kind of broadly conceived fit in with
global workspace theory and how these

ideas can kind of
take help make sense of some puzzles
within global workspace theory and also
kind of give
predictive processing a theory of
consciousness as it were
um and then in terms of
i published a brief paper kind of
building on jacob's work
where i kind of said hey he is that's
where the predictive global learner
workspace comes from
um or that term which is just a term
that i kind of want to introduce just to
denote basically kind of predictive
processing approaches to a global
workspace style architecture
and so when ryan and i started chatting
because of that paper
we really our goal was to put
some fairly vague idea conceptual ideas
into a more
computationally precise framework and
that's what we did here
um and so the idea was really just to
use an active a
fairly straightforward deep temporal uh
partially observable markov decision
process
solved via active inference and show how
by kind of tuning some of the parameters
you can get basically all of the global
work a very large perception of the
global workspace literature and all of
those empirical results
kind of just falling straight out um
of the model and what's i think really
important

to me to my background is in cognitive
science and philosophy
um and so i did my undergrad in
cognitive science and did my
masters in a computational cognitive
neuroscience lab
um and so although i work on
for a long time for a while i was really
interested in visual awareness and
visual consciousness
i was around a lot of experimentalists
and it's really important to me that all
of these ideas are testable and we
actually
specify testable predictions and so
really what we were doing
in this paper was trying as best we
could to one
reproduce a bunch of existing results
but also kind of extend the model and
show where it actually gave testable
predictions
um and some of those have actually been
confirmed um recently
which is kind of nice so
um i'll just kind of flip through some
things there are just a couple things i
want to highlight
and so the first thing is just to kind
of define terms
uh so we called it
um a active inference model of visual
consciousness and that was really just a
kind of
so conscious is the way i think about at
least maybe this is some idiosyncratic
but
confidence to me is just a kind of a

catch-all term
um it isn't one thing there are lots of
sub
things within consciousness that i think
can come apart and so block kind of very
famously in 1995
proposed the term access consciousness
where this is just content
that can be reported from the first
person perspective and it is available
to kind of cognitive control processes
and working memory that's not a quote
from him that's going to be paraphrasing
but that's roughly speaking the idea
and so that allows room for there to be
other processes
or unconscious processes saying the
visual system of which you aren't aware
and that can nonetheless influence
behavior
and he then also proposes it says chief
in that paper his chief kind of worries
distinguish between access consciousness
and phenomenal consciousness where
phenomenal consciousness is kind of the
experiential aspect of this and he
thinks that they
so this is now and he thinks that they
can come apart and that's what's
controversial so
i i don't personally agree that they can
come apart but i i think it's a
it's a useful distinction nonetheless
and so i think in terms of thought
experiment i don't remember where
it might not be it might be in that
paper it might be elsewhere but he's
kind of talked about

the example at least to me was really
intuitive was if you imagine someone
with a jackhammer
or if you're doing if you're working at
a desk and
you can hear a jackhammer in the
background
you might
what at some point you'll become aware
of the jackhammer and
but it was kind of a part of your
experience all along you just weren't
aware of it
in a way um and to me that sounds rather
odd but that's kind of
but that's fine we can we can set that
aside we're just defining terms here
yeah i mean i mean obviously i agree i
mean i think there's lots of cases like
inattentional blindness and change
blindness and things like that where
where you might you might think you were
aware of something the whole time but
actually
work but anyway side thing yeah exactly
i think like there are people
i think i've forgotten who to coin this
term but it's like the refrigerator
light illusion
like you open the refrigerator and the
lights on do you think it was always on
yeah it's kind of a nice analogy
okay so just briefly about the global
workspace
so this idea kind of started in the late
80s with bernard bars and then
stan haan's group really just took it
and ran with it

and so the idea is roughly speaking
there's this prefrontal parietal network
that connects
in kind of motor processes attentional
processing
perceptual systems memory and all of
these systems are in some sense
competing
for access and they are otherwise
relatively kind of modular and
disconnected
but when they what is access to the
global workspace
or the content that kind of wins uh
gains access to global workspaces
broadcast throughout the system so it
enables kind of
otherwise isolate subsystems to share
information in a common format
and he's identified this with a
prefrontal
parietal and this like prima faces a bit
of
quite a bit of evidence for this so
across meta analysis in report paradigms
so actually people kind of like actually
report their experience in binocular
rivalry and phenomenal masking
um there is a network of fronto parietal
regions that is
active for uh
conscious but not unconscious you're cut
off we can't hear you
oh really i can actually still hear him
oh maybe it was just me then oh good
okay no worries thanks for pointing it
out though ryan
and uh yeah continue along girl yeah so

there's basically
the the long the short of it is a cross
meta analysis that actually does seem to
be quite
quite a bit of evidence for the view
that there's this frontal parietal
network
um and this story gets more complicated
when we get into no report paradigms
maybe we can chat about that at the end
or next session because that's kind of
the work that ryan and i are really kind
of concentrating on the moment um
okay i think we can so we'll i'll just
kind of talk briefly about
these two and then kind of move on to
the model
um so the idea so basically
in a cross event related to um potential
studies so this is a kind of contrast
marking study on the right on the left
and on the right there's an attentional
blink i think
goodness yes it is um basically what you
can see is that there's this red line on
the left
and there is this little p3 it's like
literally labeled with a p3b
here and there is this late kind of
seemingly non-linear all-or-nothing
event-related potential over fronto
central electrodes
um that seems to characterize reported
versus non-reported consciousness uh
content sorry uh reporter bull
rather not not um and one what this
actually this is actually
i should just flag for anyone who's kind

of actually in the neuroscience of
consciousness
who is watching this this is no longer
considered a
neural coral of consciousness and i
agree with that um
it seems to drive be driven by task
demands um but nonetheless it is
still a question why it is that when
something is conscious and task relevant
you get a p3b
and that's kind of what we tackle in the
paper
and so really all of the evidence for
the global work for global workspace
theory i think it can be
summed up quite nicely with this 2006
kind of taxonomy
and so the idea is is that you can
orthogonally manipulate
stimulus strength and attention so
similar strength you might just imagine
that either might be
like time of presentation so if you have
a classic kind of
masking experiment you'll have i don't
know it might be a gabor patch
then just after it there'll be some
static noise
and the static noise kind of masks
is the mask to the gabor patch and
depending on the type
kind of the distance between the mask
and the stimulus you can render
something
unreportable so and nonetheless it can
still in some circumstances influence
behavior

and so you can imagine that stimulus
strength might be something like
either you can you could get it might be
contrast it might be the distance
between the mask you could
do both it doesn't really matter for our
very coarse purposes here
um and then in terms of
you can also vary attention so either
have attention to the task or tension to
be distracted away
and the idea is that when there's a weak
stimul a bottom-up signal
and attention is absent you basically
just get very little activity it's
whatever activity there is it's confined
to very early striat cortex
when you have attention is present
but the signal is still weak you get
some priming
um but
it's still unavailable for reports and
so at that level starts to influence
behavior then at the kind of the
pre-conscious
level or what they call pre-conscious
borrowing a term from freud
um this is where this is this is
attention inattentional blindness
effectively so you have a very strong
stimulus the gorilla can be walking
across the room and wave at you it can
be in the dead middle of the fovea
and yet people still won't report it and
the idea is the id this is kind of
activity that's still very strong it's
within the visual system but it's kind
of gated

out of um awareness
via a lack of attention um
at least most of the time
and then when in this last quadrant here
when you're attending something
there's a strong signal you get this
widespread kind of bifurcation
in neural activity
that characterizes aware versus unaware
states
and so we kind of had or personally i
had a couple of
reasons i was there a couple of reasons
to be a bit suspicious about this and i
actually do think web workspace theory
has been updated
to kind of accommodate these but
nonetheless i'll go through them
um so basically unconscious information
can
still be decoded uh from frontal regions
so pfc activity is not synonymous with
consciousness so the idea is that you
could might have
the idea that there is just a global
workspace and you enter it
associate loosely associated pfc and
entering is sufficient for consciousness
this can't be right
um they have a solution to this they
then basically introduced a bayesian
model
of conscious report where consciousness
also requires a stimulus
representation to be separable from a
noise distribution i really like this
model
in the beginning of my master's thesis i

spent a lot of time kind of playing
around with the code it's really cool
but i it's it's it's an ideal bayesian
observer
um it doesn't really give any neural
predictions at all um
i don't know and that that's that's
that's fine limitations
it's an ideal bayesian observer um
that still would be nice if we had
something a little bit more
neurally detailed
a christopher could you actually just
unpack that slide 13 just what are the
discs representing and
how does this relate to bayesian
statistics oh yeah sure so basically
each of these axes is a source of
evidence so evidence for
stimulus y evidence for stimulus x and
then
um basically these two
colored lines the pink colored line or
the
green colored line correspond to
different decision criteria that you are
imposing on a space
so the idea is is that alternative to
alternative first choice tasks where
you're forced to choose between two
things
or versus report something as
present or absent actually employ
different decision criteria
and so you can imagine these as a
essentially a multivariate each of these
as a gaussian distribution each of these
circles as kind of

a univariate gaussian distribution
where if you are in the blue you that's
kind of
stimulus to the left if you're all the
way
to the right or on x-axis that is
stimulus on the right um where
and then this green thing kind of close
to the around the origin
is a noise distribution so this is just
saying there's nothing there and
basically what conscious or awareness is
on this account
is basically these
um distributions or
either left or right being far enough
away from the origin that they don't
overlap substantially with the noise
distribution so when you impose the
decision criteria on the space
you are confident that you have seen
something
does that kind of make sense nice thank
you
cool okay um and then seeing this is
unseen ignore
ignores the noise distribution it's just
separates left versus right
so really cool model i really like it um
but yeah
unfortunately not quite as nearly
detailed as i would like
second problem is the p3b does not equal
awareness
so these are two paradigms so it's now
being replicated
in a no report condition basically a
person

is there's really sophisticated
experimental techniques
that allow you to basically access or
gain access to a person
make inferences about what a person saw
without having to make explicit reports
and when that happens
you see that basically the p3b which was
at one point taken to be a signature of
awareness
just vanishes and that's a bit of a
problem
so we'd like an explanation of that
and then the last one isn't really a
problem so much
as just a a point of absence so
there's a really large body of evidence
showing that
expectation seems to be a pretty crucial
factor in determining
the content the visual consciousness
so across paradigms continuous flash
depression binocular rivalry motion
induced blindness blah blah blah
or there's all these areas where to show
that expectations
probably along the lines of similar to
attention
should be a part of part of any taxonomy
or any
taxonomy that claims to kind of try to
be complete of the factors underlying
conscious access
and the global neural workspace as it
stands doesn't really describe how
expectations are implemented
and so our job or what we saw
ourselves as doing really in this paper

was just to kind of
show how we could account for this
existing work
namely the taxonomy
and also explain how the p3b doesn't
equal awareness
um or show hapso show how that
dissociation arises very naturally
and three show how expectations could be
plausibly be implemented and give kind
of a
more give a model that generates
predictions about expectations i should
say
and has a story about how they're
implemented
okay um and i think the the real thing
is it's it's okay to revise a model
like revising models is
that that's just good science um
and i think one of the part
of global workspace theory that made me
a little bit suspicious
lately is that the revisions of problem
one and two
they didn't generate surplus neural
predictions really
they were just kind of they abandoned
the p3b signature of conscious access
and they event and they endorsed an
ideal bayesian
observer model of subjective reports but
neither of those things
they were kind of responses
to problems that didn't themselves
generate new predictions that go on
um and that's kind of a characteristic
of what's been called degenerative

research programs by
lactose
and so i mean that we can argue about
whether this
is a fair characterization of global
workspace theory but i think this is
kind of at least a nice way of setting
up where we were coming from
okay um would you like to give a
a primer on active inference for partial
observable markov decision processes i'm
not sure how much more you want me to
talk or whatever
um i mean i this was an hour long this
is an hour i have an hour worth of
slides
um so i could talk forever but maybe you
want me to skip to the results or
just summarize the model or what let's
actually uh
hear your take here this is just a
really awesome presentation and then
everybody here maybe let's write down a
question or two so then we can have
another however many minutes you want of
the slides have about an hour or a
little more for questions and then we'll
have next week for questions but it's
really great to get this information out
there because the way you're sharing it
is awesome
yeah cool okay well i'll i'll try and be
a little bit more detailed when i go
through the slides because i've kind of
just been clicking through things at the
moment
okay
so what just a brief this is this is a

little bit of a point of confusion and
ryan may want to kind of come up
describe this as well so
what is active inference well it's both
a noun and a verb there's a noun that's
kind of
you can use it to describe two things uh
it's a
it's just a mathematical framework that
allows one to formulate models of kind
of
scented behavior using equations that
kind of justified a priori
from a very general set of assumptions
about the nature of self-organizing
systems
it's also that you can attach a process
theory to that mathematical framework
that allows you to give predictions
about various neurophysiological
variables like firing rate of neuronal
populations erps and kind of how
neuromodulators like dopamine or
noradrenaline
affect things like firing rate in the
rps
and also behavior um it's also used as a
verb
and especially in kind of maybe pre-2016
literature that's basically what it was
it was basically an extension of
predictive coding
to motor control and so people
often use active inference just as a
catch-all for basically the predicted
coding version of motor control
and there's been all sorts of confusion
in the literature where people are

taking
kind of active inference as a theory
basically of of markov decision
processes right and they're conflating
that
which and because markov decision
processes being a decision process is
kind of a model
of bayesian decision based optimal
decision making
and then if that's conflated with a
theory of motor control or you
use a motor control account of decision
making you're gonna get
say bizarre things um don't actually
track what's going on
yeah i mean i would that would that's
that's the main point if anything that i
would want to
you know kind of come you know fully
emphasize is that like
old active inference is a theory of
motor control i think they're old right
i mean people still do it
but yeah inference where that just means
you know like hold a prior constant and
make a reflex arc
move the body to you know minimize the
prediction error with respect to that
prior about say like
you know like proprioceptive position or
something like that right
that's just a theory about once you've
decided what to do
how to get the body to move accordingly
it doesn't have anything to do with
deciding what to do
right so another control thing is just

you know
how do i move how do i get the body to
move how i want to using
something predictive coding ish right
whereas you know
we're talking about kind of like current
is is again it's a it's a theory of
decision making
um which is like it's totally different
i really think these these these
there should be two different terms
right for these for these things yeah
that's the conflation
yeah exactly thanks and stephen do you
have a quick question
um yeah i'll just be curious like as
you're here you know you're doing work
in this
paper here with the neuronal context
of how deductive
choices are made and it taps to the
phenomenological
questions as well do you think that's
when these
you see these problems happen a lot
because when when
people are you're trying to move between
different aspects
of um fields of practice which are
sometimes thought about separately you
know phenomenology
phenomenology is often talked about in
the inactivist field and you've got the
neuroscience and they're often clean and
separate but
when you need to bring them together and
try and talk about this bigger picture
the distinction your the distinctions

you're making
become even more important if that makes
sense because
there seems to be a lot of conflation
that happens with that so i'll just be
interested in your thoughts
um so i think
one thing to say is so i i
like an activism um especially what's
kind of been called computational
activism i've forgotten there was a
really awesome paper out in synthesis
um i've forgotten the author name name
of the author but sorry about that
um but to me that stuff doesn't
really have that much to do with what i
would call
consciousness or what i think of as
consciousness as like a cognitive
scientist i suppose
um when i think of awareness i think of
a very
sharp my explanatory the thing that i'm
trying to explain is very specific
it is there are these states of which
you are unaware
some of which you can become aware of
say for example
like the intentional blindness case um
or in phenomenal masking
in some cases there is a visual stimulus
in one case where it's unaware and
there's a very small change
physical change the stimulus but there's
this radical non-linear change in
whether you can report it or not i'm
interested
in that contrast between reportable

stimuli
and non-reportable stimuli now along
with that comes when stimuli are
reportable generally speaking
there is a we might also be interested
in characterizing kind of the content of
experience as it were not just the
availability for report
um so we're going to get to that
actually kind of towards the end of the
presentation we actually do endorse a
version
of the phenomenal consciousness access
consciousness distinction
but really i think
to answer just kind of repeat repeat
myself somewhat and answer the question
all that i'm trying to explain with this
model is that distinction between
con except content that's accessible and
content that's not accessible
everything else i don't really mind like
people can call it whatever they like
uh i'm not interested in modeling it
yeah so i mean i mean in general right i
mean again this just speaks to kind of
having
precise theoretical constructs and
having a good kind of toxic you know
taxonomy of these things within a
particular area
and also you know if you're going to try
to do any kind of
um interdisciplinary work right where
you have constructs in one kind of field
and you want to find some way to
map them on to right or translate them
into

what you think they ought to be right in
some other
uh you know in some other theoretical
language
right then i mean then also specifying
that in a precise way and how you kind
of recover one from the other
in terms of the way they're
operationalized empirically
right i mean it's also something that's
that you have to do right to do this
work
well you know so i mean like you know
chris already started
right to kind of just people really
loosely
you know at least not in all cases right
obviously some people are very precise
but in some cases
the word consciousness or awareness gets
kind of thrown around to mean a ton of
different things
which makes it confusing and also makes
the term
you know uh imprecise and therefore not
really all that useful
um you know for instance there's
distinctions that people make between
right what we're talking about right
like whether a particular
content particular represented content
in the brain
and if people report um experiencing it
versus if people report
not experiencing it right um but then
there's also some people talk about
levels of consciousness
right which is a totally different thing

you know like that's like
being in a coma versus being in like a
minimal you know like a vegetative state
versus um you know being like awake but
really sleepy you know versus being
really alert right that kind of thing
um you know and there's which is a
totally separate thing right i mean you
can kind of think of one as a
as a precondition in some sense for
another right like
you have to be in a level of
consciousness
that corresponds to being awake right to
have
any you know or organ or like dreaming
right
but your brain has to be in a certain
general state
of of consciousness um level wise
to be able to have the experience of one
particular content over another
right so that's can saying being awake
is a precondition
for studying the thing that we're
interested in which is why you become
conscious of one thing versus another
given very similar stimulus conditions
exactly you know so so there are i mean
there's there's more
right different distinctions right like
i mean like i said there's this you know
distinction between or or
you know potential distinction between
phenomenal consciousness
right just like the what it's like to be
experiencing
something versus this access

consciousness
this thing which is roughly being being
aware of the thing you're
experiencing um and uh
and like i said several people think
this kind of thing can come apart
um i also i'm very skeptical about that
um um you know eg the the bridge exact
the fridge flight example
um chris mentioned earlier um you know i
mean like i said at the end of at the
end of this paper
we do in some sense describe how our
model
um can can identify something like
something like two different types of
access where one type of access
um is kind of what's minimally necessary
for
empirically verifiable phenomenology
um um whereas the other is more kind of
what determines the actual reporting um
but you know so there is something like
a distinction in our model but it's not
the standard kind of phenomenal
consciousness versus that just
consciousness distinction
um but yeah but anyway i mean there's uh
my point is
there's all these different sorts of
things you know you mentioned like uh
somebody mentioned like the
like you know phenomenal like literature
where people just kind of describe
you know the way that phenomenology is
or kind of the dynamics and
phenomenology and things like that and
then also as a kind of completely

different
very kind of descriptive you know field
right that doesn't have to do with this
just thinking about
what becomes you know experienced versus
not
um so i mean the the general kind of i
mean against is already kind of like
long-winded answer but just
just uh i think very precise
terms and very precise mappings between
terms in different
fields or necessary to do this kind of
thing um
well at all and that's not always the
case all right
yeah just to briefly follow up on that
like i
really what i'm interested is this is
kind of was put out in bernard bars
he called it the goodness i might
butcher this i've gotten
basically a minimal contrast approach
where you have
some content of which you're aware some
content which you're unaware
and up to all about up to some limit
the the physical properties of the
stimulus should be almost identical
there might be 10 milliseconds extra
between the mask
and the stimulus and that's the only
thing that renders it so
then the idea there is that we can
contrast
the neuronal kind of consequences of
being aware of a stimulus versus unaware
independent of hopefully the physical

properties of the stimulus
and so so just to give an example and
just to make sure that i'm being
charitable
um so there was recently a
uh special issue of i'm fairly sure
especially issue of the journal of
consciousness studies on like
uh consciousness in plants um
and i guess one thing that i'm not
entirely convinced of
is that we're really talking about the
same thing
i think that's great and people smart we
should let smart people do what they
want
and study all of these things but i'm
not convinced for example that
the process or it will take a lot of
convincing to
convince me that something like
phenomenal conscious whatever you want
to call phenomenal consciousness in a
tree
or a plant really has anything at all
to do with this minimal contrast
approach
in like the neuroscience of
consciousness
yeah one one other one other thing again
just like things kind of popping up in
my head here um
uh
okay thanks ryan also maybe you could
talk like a little closer into your
microphone it gets a little quiet
okay thank you all right continue and
i'm really curious to see how active

inference is going to play
into this oh sorry sorry i just
remembered what i was going to say
so i think another thing that sometimes
gets plated is
is just um representing things
versus being aware of them um right so
so you know like a major so it's a for
example
right i mean in the in the act of the
inference literature there are other
papers that have described
um you know things like jacobian monism
right is something that like i'm like
one of carl's
recent papers um you know where
where more or less it's a story about
how
via the free energy principle you can
come to have systems where the internal
states of the system
um come to parameterize or track in some
way the states
of the world outside of the system
right so so in a really broad sense you
have some
some um some way in which the internal
states are representing or keeping track
of
right the external states um that as a
kind of representation
um or the kind of seeds of internal
representations of what's kind of going
on out in the world
but the idea you know with this sort of
literature is
that the brain can represent a bunch of
stuff at the same time

but only some of the stuff that's being
represented becomes conscious
at any given moment right so like i can
flash like
a word i can flash like so unlike some
studies for instance i could flash the
word guilt for you
really fast um and you won't report ever
having
seen the word guilt but you'll you'll
actually
behave in ways that are more consistent
with
uh feeling guilty or something like that
right or or i mean simpler like
perceptual or semantic priming examples
where you might flash something really
quickly again a person doesn't
um you know say they are aware of having
seen anything but
they'll say complete word stems in one
particular way versus another right if i
flash an elephant for you really fast
you don't know i see it you don't know
you see it
and i give you like a words then like el
you know and it's like complete the word
people are more likely to say
elephant than they would be if you
didn't flash it right or something like
that
right so behavior is affected by
something that was clearly
caught right by the brain but but
despite that it was represented
um you didn't become aware of it right
so i just want to make that distinction
as well but

being conscious of something is distinct
from whether it
whether the brain actually uh in some
sense knew what's there
yeah yeah okay so
um just the idea
so we used a passage over markov
decision process
as our model a hierarchical one i should
say so this is just a single level
markov decision process
and just very briefly if pomdps
i'm just going to call them mdps just
for brevity
they describe transitions among kind of
hidden unobservable variables and the
sensory data they generate so you can
have a hidden state of the world
and the observations generated by that
hidden state
and the goal of active inference is to
infer the states and the sequences
or action sequences or policies pi
um from the set of observations
and so i'll just kind of like step
through very basically how this would
work so
you could just have a very simple case
right here where
where do this is just a graphical
representation of bayesian inference
we can just have a hidden state and we
have a prior over that state
and we have some likelihood mapping
between that hidden state
and some a
state of the world i think i'm getting
feedback from someone's computer

okay try again
continue christopher thanks okay thanks
that's better
um yeah it's funny how that was how um
that messes with you we've got
predictive models of
when our when our speed speech timing
should be at least we know that you're
awake
um okay
so we've got yeah so this this comes out
to basically if you then
invert this model this is just exact
bayesian inference
now if we start moving through time
we have to add in transition
probabilities and so this just imagine
this first level of model is just a
markov chain it's just
you have some initial state you
transition into your first state then
you have state transitions that govern
how you how those states change over
time
but you also have variables but you
don't ever get to see this markov chain
all you get to see is a set of
observations and
so the idea is these observations are
generated by these hidden states
and so the task of inverting these
models
is to infer the most likely hidden state
conditioned on
some stim stimulus
um and so you get this this is this
little sigma here is a soft matte
denotes softmax function

and basically you can derive message
passing algorithms um
or using message passing algorithms
rather you can derive
very simple update rules that allow you
to come up with posterior probabilities
over states
and then actions under this framework
are then just kind of state transitions
that the agent has control over
so you can imagine in like a
psychophysics experiment
the actions this is a super limited
exp set of action space for the agent
basically they can hit left or right and
they might be able to move their eyes
so those would be the policies that they
can control everything else
the state the screen those are hidden
states but they're just not transitions
transitions between those hidden states
aren't things that the agent can control
okay so then in a way so how do we
select actions
well we select actions according to the
frenzy functional
so this has two component and you select
the action or the policy rather
that is kind of best to expected free
energy where this has two terms
the first is the expected cost
so this is kind of the kale divergence
or the difference between
our predicted outcomes which is this q
here and our
under predicted outcomes under some
policy
um so what the outcome i expect given

that i take a certain action
and the predict and just our um
and our preferred observations so these
are things like having preferences for
winning versus losing um where the agent
generally is but you can encode the
agent wants to win
or not to lose or something like that
the second term here this is the
exploratory component
this is the expected ambiguity where
this is basically
the expected entropy of the likelihood
mapping it's the idea here is that you
seek out actions or seek out states of
the world
that have a precise mapping to sensory
outcomes and so for example if you're in
a hotel room
um that you've never been in before it's
really dark
the best way best actions to take to
minimize your uncertainty is just to
turn the light on
um because that will give you a really
precise mapping between the hidden
states of the world
out there and kind of the observations
that you generate
all that they generate and so the idea
is that
expected fringe is minimized when both
of these terms are minimized
and it's actually a little bit more
complicated than this i should say so
policy selection isn't just um
the minimization of expected fringy it
also involves minimization of

variational free energy but i would just
say what's the model stream
um to get into that for our purposes in
this conversation we can just say
policies are selected that best minimize
expected free energy
and variational free energy where
variational free energy
is also basically a stand-in where the
posterior optimal posterior here
that you infer will also kind of
you minimize you you infer the optimal
posterior
by minimizing variational free energy
and
okay so what about i haven't really
talked about the brain so far
um one of the beautiful things back of
inference is that comes the process
theory
um and so belief updates those
changes to your posterior states
correspond to changes in activation
levels in response to input
and the matrix entries here d b
h c blah blah blah these
are all h actually isn't really a matrix
i guess it's a matrix entry sure what is
h
entropy
um so you can just these correspond to
synaptic connection strengths
i mean this is just like a very standard
assumption across neural network
literature really
and just in your neural science in
general where
synaptic changes in weights of synaptic

connections
uh correspond to some kind of like
function approximation
okay and so then you get these kind of
somewhat intimidating looking
equations we can just select the ones
that are really relevant so this
first one here is state prediction error
and so minimizing state prediction area
in perception
corresponds to where the corresponds to
essentially the inference of the
posterior probability
q of s whoops and
you can cast this as a gradient descent
on variational free energy and that's
actually where we get
our erps from
um christopher could you go into a
little detail what the variational
means independently or as related to the
free energy here
yeah sure so variational free energy is
just to
distinguish it from something like
thermodynamic free energy
um so like there are formal
relationships between these things i
would just say
read carls kofristen's work if you want
to know about that
but basically uh variational frangie is
the kale divergence between
a generative model and your approximate
posterior distribution and what you're
trying to do
is change the approximate posterior
distribution

uh it slowly but surely so
change it bit by bit you nudge it in the
direction of steepest descent
until it basically this error term here
which is basically the difference
between the log difference between
your generative model and your
approximate posterior
you you change s
until that difference is at the minimum
and the rate at which that changes so
this little
v dot here is your erp
so the idea and i'm going to go into
that a little bit more in a moment
um so the idea is that you have
our the log probability of our posterior
is essentially a non-normalized log
probability
it can be positive or negative like a
voltage
and then you have you can have your erps
and this is just like your change in
membrane potential
over neuronal populations and that
corresponds to roughly speaking like
these big potentials we see at the scalp
and then we have a normalized firing
rate where
um you will basically we normalize the
depolarization variable through by
branding it through a softmax function
and the softmax function is just a
generalization
of a logistic function to
more than one input more than one cl
more than two classes
and what's nice about using a

generalization of a sigmoid function
is that everyone who's looked at like
rate models of pop in popular population
nodes for example or
somewhat sometimes called main field
models um
with these basically you can have this
firing rate versus input curve where on
the x-axis you have the amount of input
that you're driving into the population
and on the y-axis you'll have the firing
rate of the population
and it's logistic function so it's going
to be bounded a roughly speaking between
0 and 1.
so what you'll end up with is
a this kind of sigmoid style looking
curve
that we treat as a probability
distribution but can also be interp
has kind of this lovely biophysically
plausible well
i don't know about plausible but
biophysically sensible
maybe interpretation as a normalized
firing rate
and that kind of allows us to translate
between this kind of normative models
thinking about probability distributions
and what the brain is actually doing
okay is everyone kind of happy with that
yes stephen do you have a quick question
just one quick question just is the
expected free energy um on policies more
important at level two
and level one it kind of can soft max
itself
um or is it that's a bit simplifying but

does does expected free energy go all
the way down or is it a bit more
you can generally have models that
uh you can have policy selection at both
levels
you can policy selection at just the
lower level or just the higher level
depends what you're trying to model
really
generally speaking in models so i in the
model we're going to be using i'm going
to introduce in just a moment
we don't have policy selection at the
first level
um we had policy selection at the second
level
but you might for example there's work
by thomas parr
and uh ryan and i kind of working on
some similar work um
as a part of my thesis where basically
the idea is that the higher
higher like working memory-ish
pre-frontal level
you have goal-directed policy selection
and at the first level you just have
kind of
epistemically guided policy selection um
and so the policies that are being
selected at the first level might be
something like
where in the visual field to point your
attention or to move your eyes
and that's just kind of determined by
the salience of various items which in
this framework
corresponds to essentially the
information gain offered by

all the precision at the first level
a matrix um so hopefully that answers
your question
yeah thanks
so as a as just a little bit of kind of
a generalization of that i mean
really and you know and of course i'll
go into this but but the way these are
this is modeled officially is just
each lower level trial is just a full
trial as in if you were modeling it
separately
the higher level is just putting a prior
on whatever the starting whatever the
starting prior is for each lower level
trial um so i mean in a sense
you're modeling these try you could be
you could think about each trial either
at the lower level of the higher level
as being modeled
independently in some sense right each
one can have a set of policies
each can have anything that a single
level model could have it's just that
the higher level
is putting fires of some kind on the
lower level
and then the posterior observations at
the lower level
are then feeding back in as the
observations for the higher level
um so so it's
just just think about them as as just
separate kind of trials
on different time scales where they're
just connected by their priors and
posteriors
yeah exactly um and so i guess what we

did here was we wanted to have something
like
to implement something global neural
workspace like
in a deep active inference architecture
and i what i want to highlight i have i
have mixed feelings about
defining the term the predictive global
neural workspace because people lump
lump this model in in a the few people
who have been kind enough to cite this
work
have lumped us in into like models of
this as being kind of just another
version of the globally on the workspace
um
i don't think that's accurate uh we took
the global narrative workspace kind of
and all of the phenomena that
that it encompasses kind of as something
that we want to explain
um both ryan and i really like the
global neural workspace like
standard harness bars huge heroes of
mine and aside from kind of like my
personal
love of the theory it's also among
kind of neuroscientists and philosophers
who work on consciousness
about 30 the majority of them which is
about 35 or something like that
in 2018 said that they regarded as the
most promising
neuroscientific theory of consciousness
and so we thought okay if we're going to
go into this literature this is what we
should be modeling i think the phenomena
encompassed by this theory is where it

should be the starting point
but with that said it's our model is an
active inference model through and
through
um and so it really does kind of stand
on its own independently of the global
around the workspace and so i just kind
of wanted to flag that
and so i i don't use those words or in
the draft of the paper that i'm writing
at the moment kind of following up on
this one i don't use those words and i
don't do it
that's very deliberate um
but anyway sorry that was an aside um
that's a very esoterical side of that
anyway um so the idea here is that we
implement something global in the
workspace like in a deep
deep temporal architecture so this
second level
we identify deep level we identify with
the workspace
and so as ryan was just talking about
you have these states the second level
which are really like beliefs about how
trials at the first level evolve over
time
and so they provide a prior at each time
step
of the lower level they provide a prior
over the hidden state and then
after when there's posterior inference
at the first level after that's happened
it then shoots back up to the second
level
and that acts as an observation for the
second level

okay so we the functions we tip the idea
here is that the functions we typically
associate with conscious processing
take place at the level of the hierarchy
that's kind of
deep and temporally deep enough to
abstract away from processes that are
entrained by the moment-to-moment
sensory flux
and this allows the agent to kind of
construct a report of its experience
or to do anything any other kind of kind
of quite abstract cognitive task
and the idea is that ignition occurs
when a first level state is in third by
the second level
so ignition sorry is basically like what
dehan has called
this non-linear bifurcation of this
fronto parietal network
in the neuronal activity so the idea is
that ignition for our model occurs when
the first level state is inferred by the
second level
with a high enough posterior probability
to kind of influence policy selection
at the second level
oops so okay on to the task
so this is a very kind of general task
structure which we borrowed from a
really awesome empirical study done by
michael pitts
in 2014 and colleagues um so idea was at
the first time step
there was a forward mask presented and
so
second time step there's a target
stimulus so these little lines in the

center can rearrange themselves into
a square and the second type there's a
backwards mask of the third time step
and that the agent is then asked to
construct a report of its experience so
whether it say whether it's or something
or not
um and the idea roughly speaking these
little
colored circles on the outside this is
how we manipulate attention or and how
michael pits manipulate attention
um and so they either had people attend
to the inside and perform a task
basically click a button whenever the
circle change the square changed or
click a button whenever the circle
changed
um and so they manipulated kind of
what was task relevant and i'll get into
how they manipulate awareness in a
moment
but roughly speaking the way we
implemented the model this is kind of a
bayes net
representation of the model and then if
we just kind of click through it
at time one there is a forward mask
presented and this just corresponds to a
black
with a square trials that's the full
this is kind of a
belief at the second level there is a
full sequence this is beliefs about the
full sequence over the full trial
so this is a time step one um there's a
report hidden state factor here
it's in weight because we haven't asked

it to construct a report yet
then at the first level each of these
second level states kind of implies
a first level state that makes sense and
each of these first level states kind of
implies
a observation and
attention in our model was basically we
just hard coded in
as a but we didn't have to hard code it
you can get these things to emerge we
just
it's just a little bit easier in our
case to do it this way we basically hard
coded in
so that attention corresponded to
modulations of the
mapping of the precision of the mapping
between these first level hidden states
kind of whether there was a square or
lines out there in the world and the
observations
so okay so first first time point
forwards mask
we see a sequence type is
going to be a square there's a trial
phase one
report okay next one we see trial is
moved to two
we've now seen the target has now
rearranged itself into a square
we did this for two time steps and we
did the reason we did it for represent
the target stimulus the target stimulus
the two time steps is to allow recurrent
feedback processes from the first level
kind of first level second level to kind
of recognize what's going on then feed

back
on to the first level kind of give
that's
that was kind of our way of modeling
this ignition style process
then fourth time step we take away the
math we replace the little square of
just the mask again
and then from five to eight
do we allow the model to construct a
report of its experience so basically
this scene hidden state here implies a
sequence of
language processing states at the first
level
i submit a square yeah just it says i
see a square and then what would it say
if it was unseen
uh i didn't say anything
i mean to be fit just to be clear like
uh
we could have and the model would be
exactly the same in
all aspects we could just chop off this
kind of language processing part of the
model
and the model would behave exactly the
same way
and we could still use it to generate
reports and all of that stuff
what we wanted to do was really just
kind of use this as kind of an intuition
pump
for what having this deep temporal
representation
allows models to do so holding something
at that level
allows sequences or a cascade of hidden

states
to kind of evolve at lower levels that
might be something like constructing a
sentence
with a goal in mind or it might be
something like um
i don't know make a cup of coffee where
you have to hold multiple tasks in mind
and break things down
the sub components blah blah blah yep as
opposed to mapping to button that says
yes or no
if it's just a two button option then
it's a one time step paradigm
yeah exactly i mean we could do that i
mean it'd be
completely identical um this is just
kind of a
a nice way of showing what these models
can do really yeah i mean so this was
this was partially for uh for a kind of
to make a theoretical point right you
could say
right we're showing a two-level model
here but you know
the brain probably has you know i don't
know how many levels but a very large
number of levels right not just
two um so you know you could
you could kind of level the concern or
question at us that like
you know why is it that this second
level is the one that corresponds to
conscious access
right as opposed to just any other level
that's above some level
right in the brain since there's a big
hierarchy in the brain

you know so the the idea here is is that
you know we're appealing to a certain
level of
deep temporal structure right like
um you can you can flash a stimulus at
somebody
and they can see it over a very short
time scale right of like
you know half a second or something
really fast but
actually but the sorts of cognitive
processes
that can integrate that together
right and generate you know a set of
thoughts right that corresponds to being
aware of that right constructing
a um yeah constructing a sentence or any
of the things that
chris mentioned um those are much
deeper slower integrative cognitive
processes
right so the idea is that there is a
level in the brain
that is integrative enough and that
represents things over a deep enough
temporal scale
that it has the minimal resources
to um integrate things together hold
them in mind
um generate sequences of the sequences
of words that would be necessary
to communicate that you saw something or
not right so
it's that abstract deep temporal
structure
that is the necessary conditions for
what you need to get
empirically um confirmable awareness

yeah
okay and just some brief kind of
modeling notes about what we did
so to model attention and stimulus
strength we basically just altered the
precision of the air first level a
matrix mapping by passing it through a
softmax function
twice where one of the softmax
temperature
so when you pass something through a
softmax function you have a temperature
on it and the temperature decides how
precise it is and we just had a
different temperature
for each iteration of the softmax
function
and so the idea was then we had an
ordering basically of precision
where the most precise was uh
attention present stimulus strength high
the second was attention
um attention absence similar strength
high third was
you can imagine how that goes on and
kind of the reason this is
this sounds pretty abstract and a bit
kind of cooked up and that's true to an
extent
but i think this does actually really
correspond to so if you look at
something like
divisive normalization in vision
where a tension basically is a
you will have some tuning ker neuronal
tuning curve and
basically basically like how precise
that tuning curve is

is modulated by both the contrast of the
stimulus
and then the contrast the stimulus gets
fed into
some attentional process where that will
together determine kind of the tuning
curve
that was a very kind of rough and ready
explanation of
um tuning curves but whatever um and
then
our threshold for report we basically
just set up the model to give
have a preference for receiving correct
versus incorrect feedback
at the final time step and oaf behavior
was basically modeled by simply
decreasing the preference for being
incorrect to encourage guessing
um and so i think this was just kind of
our first pass at doing report
i don't think this is how the brain does
report
um we actually in our follow-up model
we've got a much nicer way of doing this
but i think it was it worked in the
sense that it was able to kind of quite
accurately capture a number of empirical
findings so that's kind of good enough
for me at this point but this first
first model okay sorry
that was a lot of preamble um we can get
into the results
so foundational simulation um this was
just like kind of full
like a very high precision stimulus was
presented the model
kind of with either on a square present

or a square absent
trial and this just kind of to test it
um so great uh when there was a square
present the model reported seeing it 100
of the time
you see this nice kind of bump at the
second level at the first level where
um the firing rates for square increase
when we present it goes down
and chris let's just make sure that
people understand
and how to interpret these things
so just just so people understand so
um time in these plots goes from left to
right
and the grayscale from white to black
black
corresponds to 100 probability and white
corresponds to zero
um probability and the gray so is
something like
you know 50 50. um so
and these are you know cast in terms of
firing rates here so darker than
implies a higher firing rate um so the
top plot
on the left that says second level
firing rates
that just means it first sees the black
circles on the outside
so it's confident that it's either
um gonna be a the sequence that has
black circles in black square or the the
second row
the other sequence which would be black
circles and just lines
with no square um so basically once it
sees the black circles

then it knows okay this could either be
a sequence where there's going to be a
black square or when there's not going
to be a black square
so then at the and that and you can tell
what the stimulus was because if you
look at the first level firing rates
so again the next kind of plot down the
bottom
[Music]
row there corresponds to seeing the
lines and the top one corresponds to
seeing the square
so basically what happens here is it
starts out just seeing the lines
and then at the second time step there
you can see it kind of bumps up
and the agent's really confident that
it's seeing a square
which at the second level corresponds to
becoming
fully confident that it's going to be
the sequence
of black circle and square
and then the stimulus goes away at time
point four which you can see at the
first level and then stays again
just lines for the rest of it um and
then the the bottom two plots
um show the actual outcomes um where
again
black is the agent's sort of competence
in what it did
and the um cyan dots
correspond to the actual ground truth um
so that's just showing
um that the agent waited
for the first three time steps and then

recorded scene
um and kept that scene while it
generated the sequence of words
at the following four time steps and
then the right one
just kind of shows it started out silent
for the first four time steps
and then um the words that it observed
itself report were i see
a square which is just those the dots
moving down the diagonal to the right
um so that's that's just just so people
know how to interpret you know
what these plots mean yeah exactly
another clarifying point would just be
that the same simulation
is leading to all of these behavioral
outcomes and
neural correlate predictions and a bunch
of other things so it's like one
integrated model and
the model stream is kind of where they
built it up yeah exactly
um okay so we kind of talked through all
that
okay so in terms of the taxonomy here
what we wanted to do
was basically recreate that taxonomy
that we talked about earlier
so this was uh modulating signal
strength from high to low and from kind
of
and whether attention was present or
absent and then looking at the
corresponding firing rates
and looking at like the percent seen and
percent correct
that kind of thing also sensing report

behavior and percent correct which
is false choice behavior
okay so kind of couple of things to
highlight
firing rate first level is enhanced kind
of most strongly
in the conscious quadrant that's nice
that's again what we see in empirical
findings
um high firing rate for kind of square
sequence at the second level even when
reports are low this is kind of nice
right because i talked about earlier you
can have kind of
a firing rate at the set you can have
kind of activity in prefrontal regions
that is generally speaking below the
threshold for report
um that is nevertheless
oh sorry when i say below the threshold
for report report
frequency of reports are relatively low
but you can still have a preference
for um you can still have a higher
firing rate
or the representation of showing that
there's a representation of the stimulus
basically um and then the last
thing to highlight is this that the
highest firing rate for the kind of the
square sequence at the second is at the
second level
is in the conscious quadrant which we
kind of might think about in terms of
ignition
okay so then erps we see the real thing
to highlight here
is that we see a large p3 like

event-related potential in the conscious
quadrant again
like a lot of the empirical findings in
the report literature
whereas first level first level erps
seem to be fairly unmodulated
by um
by conscious access essentially which is
again what we see
so now to a very that was kind of a very
abstract kind of taxonomy we were just
manipulating
factors what we wanted to hear was show
take a concrete empirical result
and show how we could basically use
literally exactly the same model
to reproduce it so this is a really
clever paradigm by michael pitts i
talked about before and so the idea is
that you will have
these kind of uh
three phases to an experiment in phase
one
the participant is just told to attend
to the dots on the outside of the screen
and hit a button
when they dim and what the what
participant isn't told is that every now
and again
these uh things these lines in the
middle will rearrange themselves into a
square pattern
then at the end of the first phase she
seek like 300 trials
and at the end of the first phase they
are then kind of given a debrief and
they asked all right so how many people
saw a square pat in the middle and

roughly speaking it's about 50
and this is now actually really well
replicated result uh across kind of
different varieties of intentional
blindness roughly speaking about 50
of people don't report seeing it when
it's not task relevant
so that's nice so then they then all of
those
so then they have 50 of people who
didn't report seeing it
and then in fact are now aware of it and
then phase two
they do exactly the same tasks they're
attending to the disks on the outside
the outside
uh so it's irrelevant um
still task relevant but they're now all
aware of it and the debris for the end
of the second phase
everyone reports seeing it and so now
they've got this contrast
between aware versus unaware so phase
one versus phase two
where the only thing that's different
between them is whether you're aware of
the stimulus
and and and the stimulus in the middle
is task relevant and what you can see
there's no p3b there's no
late positive centro positive central um
component in the erps then the final
phase
aware task the kind of the aware phase
so all the participants are aware but
they now make a task relevant to have
them basically attend to
the dot to the squares in the middle and

when you see there is this big p3b
very noticeable okay
so just kind of some um this slide just
summarizes what i just said
then great so basically what we did was
we just used a different inverse
temperature parameter to
modulate to kind of represent each state
so phase one
very low um temperature parameter for
attention
attention blur attention
temperature parameter um that was kind
of correspond and so this is you can
kind of see the precision of the a
matrix here we bump up for phase two we
bump up
the inverse temperature parameter just
very slightly um this is kind of roughly
speaking to chorus
correspond to like a diffuse attention
so not you're not particularly attending
to one thing
one particular feature of the stimulus
you're attending just kind of diffusely
to the whole screen
and then the last state we basically
model the tension make task relevant by
making a very high or very precise
temperature parameter where you
basically get almost something that
looks like a likelihood
like a identity mapping
okay and so these are the erps that we
the erps and the firing rates do you get
out of it
and what you this is my favorite slide
in the whole plot uh oh in the whole

paper um
what you see is basically results that
look very very like
shockingly similar to the empirical
results especially for a model that is
so incredibly simple um like ours is
relatively at least in terms of
biophysical detail
um and so what you see is at the first
time step so there's basically no erp so
first phase basically no ips
and so we we give the model feedback on
every trial but what you should see is
it's a little bit darker for the square
for the um
until we give it feedback and tell it no
you were wrong
um you did we did actually present
squares
um and the model changes its mind it was
fairly sure
that it had seen lines then
the second one we bump up the attempt we
bump up the
tension parameter and now the model sees
things and it gets
or it reports seeing the
stimulus so at this so i should say in
phase one our model saw
um the stimulus 49 of the time basically
identified the empirical result was a
fifty percent
phase two it's sort of 99 of the time
basically the same as the empirical
results report phase two is 100
and then phase three it's a hundred
percent of the time um
and there there's little but by the way

i should say what the way i generated
these results is basically
i ran several hundred tr i basically set
up just ran a for loop over the stimulus
over the code and for each condition
and had it generate i think so
each kind of iteration of the for loop
generated 300 trials
and i then averaged over maybe 10
iterations of that for loop
so um each with a different random seed
and so what you should see with the
sorry that was a quick side
the crucial thing is that in this last
point there is this when there's a very
precise stimulus mapping at the first
level
there is this abrupt change in firing
rates at the second level
and that abrupt change corresponds to
the appearance or the emergence of the
p3b where the p3 bit where
remember that erps are generated they
are the rate of change
of posterior beliefs and so the model
suddenly becomes very confident that
saurus square and so you see this big
erp boost in the p3 which corresponds to
a very precise
input mapping that comes along with
something with a stimulus being task
relevant
and so there we've kind of got this
really nice explanation
of this kind of somewhat
counterintuitive result
about the dissociation of the p3 and
awareness

okay and then the last thing i want
we're going to touch on is just kind of
we then decide to extend the taxonomy
so basically this is exactly the same
but we also now manipulated
whether a prior was consistent with a
valid
expectation so we gave the model kind of
we made it
roughly speaking two times the model
photos roughly speaking two times more
likely to see a stimulus
than uh see a square then lines
um and or two times more likely to see
lines than squares
and then we presented it with squares
and so that's kind of the consistent
inconsistent prior
and then the key result is that across
all of this expectations boost
the effect of feedback on firing rate
which is nice
and in terms of erps what we see
is something this very specific
prediction
when there is so these are kind of like
fairly small differences but they are
differences um
only when attention is present do you
really see
a p3 that's again in line with the
empirical literature
but crucially when there's a consistent
prior or a valid
phase to say a valid expectation you
should see a reduced p3b
in contrast to when there's kind of flat
priors and when there is an inconsistent

prior and this is actually something
that has been empirically that has been
confirmed recently
again this was something michael
pitcher's on this paper if anyone's
interested i can kind of post a link to
the paper
and that that's really nice we didn't
plan for that to happen um
okay it is slightly annoying i would
have liked to i would have liked to cite
it
in the paper but the paper actually came
out before ours was accepted for
publication
oh sorry uh after hours was accepted
publication
rather that's what i meant to say um
okay and just a very quick summary
summary of the results
with really what is a really kind of
like surprisingly simple model
we can reproduce a very wide range of
canonical results
um from minimal contrast paradigms
including kind of fmri and erp
findings so this includes we can account
for
unconscious pfc activation we can
account for the association of the p3 in
conscious
access um where the p3 roughly speaking
reflects the velocity
of conscious working memory updates i
kind of like that phrase
and four we have an explicit and
formally defined role for prior
expectations and the implementation of

process theory
and then we've got a number of key
predictions that emerge from this
so p3 is attention dependent and invert
and should its amplitude should be
inversely proportional
to expectation feedback from fronto
parietal regions
this is a very specific prediction
feedback from frontosharinal regions
should be enhanced unconscious trials
and it should disinhibit granular layers
in the relevant sensory
population sensory cortex that's a very
specific prediction that falls out of
the process theory
that that can be tested with dcm and
then consistent expectate expectation
again
consistent expectations should
disinhibit granule layers and sensory
cortices
while inconsistent price should inhibit
them
okay so any questions at this point
christopher thanks for this awesome
presentation
do any of the panelists um maybe want to
queue up a question
how about you do this last slide with a
bat leave on that sort of
phenomenological
no if you want to otherwise we can
return to the panel
yeah perfect yep sounds good so all the
panel prepare a question or a thought
and then raise your hand when you're
ready

and give a last thought here thanks
christopher okay so
just like a brief aside on phenomenal
consciousness so
we associate that so i'll just back up a
little bit sorry
there might be a puzzle in that we
associate global broadcast and the
availability of information for report
with this second template deep level
our experience although it's kind of
like there is a depth to it
in the sense that experience is kind of
smeared over time
right there is an integration window
that's relevant for experience
it certainly isn't doesn't occur at the
level of whole sequences
and so there's a puzzle here so where
does experience live as it were or where
is the content of experience
and kind of brian i've been chatting
about this and this is the view is still
evolving but broadly speaking
i think what's right about this or what
what we should say is that the first
level states
that what's what the content that we are
conscious of are the first level states
that have a high enough precision
um either through kind of bottom-up
precision or because there's a really
precise prior that comes down from the
second level a matrix
um so basically the message that's being
passed back up to the second level
has a high enough precision to influence
policy selection or to pass the

threshold the posterior belief threshold
for conscious access and report and so
this kind of has the nice consequence
that it keeps experience at the relevant
level
so it's updated at each time step when
we get new stimulus in
stimuli in and yet it has a distinct
kind of
link to report and behavior
yeah so i mean just to expand on
expanding that a little bit i mean the
kind of idea is just that
as opposed to and again chris partially
said this but just kind of reiterate
um the idea is that um
what you consciously experience um
is is a consequence of both the
posterior states at the first level
and whatever the second level likelihood
is right so it's a function of both the
posterior there
s1 and what the structure is of a2
there and you know way to think about
that is
if you were to rearrange a2 so the
mapping was different
um and s1 had the same posterior
then that would update the beliefs at
level two in a different way
right so you know if i arrange a2 one
way
even though s1 there is still sensitive
to the same stimulus
um then if i rearrange a2 one way then
the posterior at s1
is going to lead to um awareness
of one thing whereas if a2 is set up a

different way
then the second level will gain
awareness of a different thing
right so so in some way the phenomenal
content that you become aware of
depends on the nature of this link
between
what messages the way that the messages
at the second level are kind of decoded
in a sense right what the second level
takes to be the meaning
of the messages that get passed off from
the first level
um so so this kind of nice thing like
chris mentioned where
it's the updates um that the first level
gives to the second level at each time
point so phenomenology kind of stays
at the temporal scale of experience but
uh phenomenal content still depends on a
certain type of access right it depends
on
the first level having a specific type
of influence on the second level
brian thanks a lot it's really
interesting so let's return to the guest
the panel mode
christopher awesome presentation and the
first question is going to be from
dave and then anyone else can raise
their hand
so dave go ahead my question is about
possible involvement of the air toss the
reticulothalamic
activating system either in your data or
the data that
are uh drawn on neurological data in the
other

relevant studies is there any
representation of possible contributions
by the air toss by the midbrain
and is there any accommodation in your
model
of something maybe not explicit with the
air trust but
uh data uh that
might originate there say the endogenous
attention
um so
subcortical structures are really
interesting and as for the reticular
thymic activating system
i'm curious did you did you get that
from this book by bernard barz
so a cognitive theory of consciousness
where he first introduced
global workspace theory and he has a
whole chapter devoted to
the reticulophalamic activating system
oh that's great i will definitely have
to get that no
actually i got that this from uh the
work
that uh yak punk sepp and uh mark
solms did especially the 2013
thesis of the the conscious id
he is of course a very close
collaborator with carl frist and he just
brought out
a full-length book on
the conscious id especially as
um active inference interacts with that
so so my my actual main focus of
research is
on um emotions um you know so applying
computational models to emotions

and um i'll just say that i am i have
very strong
disagreements with the um the kind of
theory proposed by jack and mark
we actually published kind of a joint
like on paper debate
where me and um a colleague richard lane
kind of
debated back and forth with jack and
mark about the right way to think
of um the way consciousness works and
i'll just say that
you know their view places this really
strong role of
affect where they they more or less say
that these midbrain
you know structures um like the pag
right the um
gray um send these afferent signals up
to kind of activate the rest of the
brain and that somehow
permeates it with this kind of necessary
emotion
um and that's kind of primary um whereas
whereas kind of the way that um the way
that we've
the kind of structure of the sort of
thing that we've defended is just that
no
like the global broadcasting ish kind of
thing
applies to all types of information
right so if you have a representation
of your beliefs about your emotional
state that has to be attended to
and sort of sent into the workspace just
like any other stimulus
for you to become aware of it um and so

i mean these sorts of midbrain
structures they do a lot of really
interesting like interreceptive and
visceral regulation processes
um and you know they do have this sort
of general
like upward modulation rule while they
well they kind of
you know my view is they kind of keep
the cortex in general
in a state where it's capable of of
representing things being sensitive to
simulate holding different states active
and other states not so it it
essentially it allows things to be
represented
in general and then to become
consciously accessible
um but you know with respect to um
the way that kind of emotions feed in
and whether the uh
the midbrain structures actually
contribute content in and of themselves
we just have theoretical disagreements
about that can i
also briefly follow up on that so
the contrast the reason i think that
fronto-parietal regions in cortex
are so important is because
there are a huge just amount of
empirical literature now in neuroimaging
and
also very recently data from kind of
invasive neurophysiology
in monkeys and most recently in mice
showing that yes when there is this
contrast between
conscious versus unconscious these are

the things that differentially modulated
now you can't there are some very
impressive
findings very recently coming from
matthew larkham's group
showing basically that you can wake
creatures up from general anesthesia
by stimulating the medial dorsal nucleus
and the thalamus
and there are these crucial connect like
reciprocal connections between layer
five of the cortex
and the medial dorso and the thalamus
and when those goes out basically uh
general
they go out under general anesthesia and
when you are awake
or when like whiskers are stimulated in
barrel cortex
when they report see when the mouse kind
of reports as in it says its whiskers
were stimulated
this circuit goes off essentially so i
think the thalamus is crucial
but uh kind of allah dehaan and every
everyone else
i think the reticulophamic activating
system and all those sort of cortical
structures
these are background conditions similar
to what ryan said it puts the cortex in
kind of the state where it could be
aware of something
but the actual contrast between
something being broadcast or
broadcast first not broadcast or being
reported but not reported
does not rely on those structures um and

i would just want to see
from those authors i i would actually
just want to see them engage with
something like
the visual awareness literature and show
that those structures are differentially
involved
thanks for that uh steven go for it and
then dean
blue oh actually then scott then dean or
blue if you want but stephen with a
question
then scott
yes so thanks um just a question
you you just to be clear so effectively
you
you tagged the degree in prediction
confidence and that mapped against the
the data that's empirical
if i'm writing understanding so you like
as
the model showed a difference in
confidence that was
showing a correlation to how firing
happens
and i'll just be curious to know if
um you think that this this sort of
key level or something where this firing
or deductive
piece can happen um where there's this
is that
where there's a shift between and maybe
a more
markovian ergodic process to something
more semi-markovian um
and less ergodic and you so that an
organism can make
non-ergodic choices in conscious

awareness
uh so i mean a couple of things so i'm
not sure about the ergodicity comment so
i think you can probably have i'm not
sure how semi-markovian systems relate
to ergodicity i
assume that you can have a
semi-markovian system in a
godix system that doesn't like that
wouldn't
surprise me um yeah so basically when
you move from uh
i just don't know actually just to flag
that but if you
i'm not particularly up on a lot of the
physics cisco physics that relates to
how these models work
[Applause]
but yes when you move from a first level
model to a second level model
they are sammy markovian and can you
just maybe define that a little more
what does that mean to you
what does that enable and maybe that
will get at what stephen was asking too
um basically the model is no longer just
purely dependent upon the previous state
at the first level of the model um there
is more information
that is determined at the second level
than what is just goes on at each time
step of the first level
that's basically it uh yeah so i think
we we have a sentence about this
somewhere yes uh you
what what the temporally deep scale buys
you is essentially
what we were saying before all of those

cognitive actions
that allow you to
do things like construct reports make
plans
etc etc and kind of abstract away from
moment by moment sensory flux
all of these things depend upon there
being a second level
and that might be reflected in the brain
there's some from the ground yeah yeah
yeah
so smooth uh
no it's a great grand depressor granular
cortex is generally speaking like
so the cortex has six layers roughly
speaking
across and as you kind of move
up or move towards the center of some
centrifugal hierarchy
depending on who you talk to and what
kind of anatomical maps you believe that
type of thing
layer four of that the granular cortex
which is essentially the input layer
from the thalamus and the feed forward
layer
that gets smaller and smaller and
smaller until when you're at kind of the
center of this hierarchy or this
centrifuge
um you essentially have no granular
cortex at all
uh this is kind of ties in lisafeld and
barrett's done a lot of stuff with this
i have problems with that view to be
honest they kind of say the top of the
hierarchy
is where they think consciousness lives

i have a rather vehement disagreement in
that i just think there's absolutely no
evidence that that's true
in terms of neuroimaging evidence uh the
areas of the relevant seem to be these
fronto parietal regions
uh they do have a layer they do have a
somewhat well-defined layer for
it's not completely a granular it's not
granular
somewhere in between um yeah i don't
know i think they
what one thing to say about i know ryan
has very specific views about this and i
should say that i really love
lisa from barrett's work i think she's
wrong about this um
i i i just want to go with kind of the
empirical contrasts right
and not speculate too much about very
specific implementation details
so look they kind of go from a very
loose model of kind of predictive coding
and then map it to these like
cellular uh kind of details about the
structure of neurons um
i'm more interested in going from these
computational properties
to basically what would measure the
level of neural imaging because i think
that's what is most tractable
experimentally
and then if we're and there's still a
lot of debates about that neuroimaging
level right
if we kind of hammer it out and decide
what's going on
at the neuro imaging level of analysis

then good
then then we can start having more
serious discussions about
particular invo involvement particular
types of cortex but i don't think we're
there yet empirically
yeah i mean i i would say yeah just to
kind of echo that a little bit
um you know it's it's important to
realize
that um it's not as though
you know the particular sets of you know
little node neurons and synaptic
connections
um in the kind of like you know
hypothetical
columns um you know that chris showed
earlier in terms of the neural process
theory
um that's just kind of one example of a
way you could set it up
there's there's a very very large number
of ways that you could connect up
neurons together in different structures
that would
implement basically the same algorithm
and in addition to that there are
different message passing algorithms
that you can use
um to solve the same graphs um
an active inference isn't defined by
some particular message passing
algorithm
right like thomas parr has shown this
you know specifically in a paper where
you can have different neural
implementations
that require more or less

neuronal resources to implement it's
kind of a trade-off between
efficiency and accuracy
where you know for instance like really
really accurate
um message passing algorithms like
belief propagation
they require more neurons to do um but
they're a little more accurate
and the brain could use that right or
the brain could use variational message
passing
which is not quite as good of an
approximation but
requires less resources to do um so the
point being is is that
it's not really that strong of
commitment to that level of kind of
circuit detail
in active inference it's more just a
theory of you can have this pattern of
synaptic connections
that implement the matrices and we
represent
posterior beliefs over states with these
firing rate functions
um and then you take the rate of change
in those to put the brps
um but so this is this is kind of the
the issue with
it doesn't really make sense to me
anyway at this
point against echoing with what said um
to start with hey let's assume that this
one
very specific way of setting up neurons
to do this
is the right one you know let's just

assume that
and then figure out what must be true
about
different areas of cortex based on their
uh
layer based on their cortical column
layer structure
um so so then it's definitely
i mean it's super interesting and
probably has um
you know a really meaningful interesting
mapping between the differences between
granular and
granular and a granular cortices um with
respect to computational function
um but um but again you know like i
don't think
um until we've really nailed down that
there is this one particular circuit
implementation
um it doesn't really make sense um in my
opinion anyway to kind of infer from
that direction
um to function um
so i should say yeah just to kind of
echo that i think
if you work in your imaging it doesn't
make sense if you worked in mouse
as a in a mouse uh model and you were
literally testing
hypotheses at the circuit level then it
would make sense something to do
but well right but in that case but in
that case you're literally
testing like the predictions of specific
message passing algorithms
yeah exactly so that was kind of testing
that's testing for message passing

algorithms that's not
assuming a particular one and inferring
from that
so anyway i mean but i mean if you i
mean for people who are interested i
mean
i mean we you know so so i and some pre
a couple previous papers have um
you know kind of built um multi-level
active inference models
similar to this but specifically about
um emotional awareness
um you know the the sort of show how you
can get
um you know from simple kind of um
lack of awareness of single emotional
states all the way up to being aware of
feeling multiple emotions at the same
time and being able to report them and
things like that
that kind of show how a structure not
too different from this one
can can generalize to um to emotions or
really to
to any other sort of thing that you
could experience
and be aware of um whether that's coming
from
inferring things about your own kind of
bodily state and its emotional meaning
from interception or whether it's
something like vision
so i mean we we haven't covered that
stuff in
any of these sorts of things before but
but um that stuff is also out there um
that might
kind of help at least show what kind of

the view on our end is about how active
inference can
uh account for um emotional phenomena
and be consistent with
the current literature on emotional
awareness as well
thanks for these awesome answers scott
with a question and then i'll ask one
from the chat and then dean or blue
or someone else
there we go um thanks great and
fascinating presentation
i was wondering this book
the intelligent movement machine
yeah and um it's it it talks generally
about the
um cortical mapping being predicted
by the movement repertoire in the motor
cortex and kind of that embodiment
kind of element one of the things i
wanted to ask about the reason
i think it follows from that this most
immediate prior part of the conversation
is when we have an active inference
setting and that model
is being employed for multiple
um perception and i won't use the
phrasing
right because my active inference terms
aren't all up to stuff but
when the inputs into the
individual who's performing the active
inference that system
are from various
sensory inputs and various temporal
distance
so there's several inputs and then
they're being synthesized

into a an action or a
an externality something's going to
reach out and again my apologies for not
internalizing all the definitionals um
what is what are some of the cortical
mappings or have you found cortical
mappings that embody
that process of synthesizing multiple
inputs
into a single action what are the core
what are the structural correlates of
the process of synthesis
of multiple different strands of active
inference into a single
action potential again i'm using the
wrong words but
nice questions this is a question
essentially about multimodal integration
yes so yeah yeah um so i guess
i i think active inference is probably
you might be able to build a specific
so i think there are two parts this
question that's an interesting question
um
first part is i'll just kind of answer
the active inference end of it
uh the way i think about active
inference at least is just as a really
useful modeling
and mathematical framework and i think
it's possible to build multiple active
inference models
of any phenomena so
i would want to hammer out an empirical
paradigm
um and then figure out okay what models
could
i build of this so that's kind of the

first part the second part is about
multi-modal integration are there kind
of
places that house moji mode i haven't
looked at modi modal integration
from since i was an undergrad i'm just
trying to remember like my
third year cog neuro courses i think
there are places in like posterior
parietal cortex
where there is integration between
multiple modal like v
you might say like vision and one other
thing or whatever it is and then that
gets shot off to motor cortex
implemented in actions um so i don't
know i think you i would just say look
at the motor planning literature
um there's lots of stuff about
uh bayesian um visual motor integration
and one and just follow up one of the
things that
that leads to is i was wondering about
the situated cognition opportunities for
those syntheses
so for instance you have um hysteresis
or my son used to say stereotypes are
real time saver
he said it as a joke right so the idea
is you have externalities that
facilitate the synthesis
so they're actually not cortical
structures they're social and
linguistic and rhetorical structures
right i mean is that could those what
i'm where i'm exploring is could those
be described
with some of the same models as the

cortical structures
so you're actually able to have a scale
independent
description of the processes both in a
social and a cortical
cognitive context for the synthesis
of multiple multimodal inputs
so i'm not sure about the scale
independent part yeah
yeah i i i hear what you're you know
scott it's a it's a beautiful question i
see where you're coming from because
we've been talking about the scale-free
formulisms
with crisp fields and others and then
this whole nested markov blankets like
you could have the neuron
and then the neural you could go within
a model and just like you said
um your model doesn't presuppose that
awareness is actually at a higher level
you're just saying there is a structure
that's deep temporal structure and
that's the state that's the timeline
that you are at but
there could be another level and then
mechanistically or functionally how that
level plays out
is going to come back to a task specific
measurement which is and a plurality of
models
so still good points i mean part of part
of i mean i think you know timing to go
back to i mean there's lots of like nice
opportunities right to build active
inference models would be very simple
you know for like multiple multi-modal
or a cross modal

right integration task you know like for
instance like um i haven't seen this
done um but maybe i have to look back
but things like the double flash
illusion or the rubber hand illusion
or you know any any of these kinds of
yeah multi-modal things i mean like the
like the double flash illusion is really
nice right because
i mean more or less for people who are
familiar um
you either play one or two tones
and they are coincident with a single
flash of light
um and it just turns out that when you
play
two tones really quick then people
perceived two flashes of light when
there was really only one
um and the reason for that is so there's
just this kind of like friar
in the mix presumably that coincident
you know things that are coincident
temporarily have the same hidden cause
but also at the same time um the
temporal resolution
of audition is just better um
this is more trustable than the temporal
resolution of vision
um so so you can just say look like the
system
just trusts right it has beliefs that
the auditory system has higher precision
and that there's a high probability of
common causes of coincident
input therefore there must have been two
flashes
um given that there was two tones um you

know
i mean that's just one example of like a
really simple well-known empirical task
of cross-modal integration that would be
pretty
easy i mean probably without a lot of
without even a ton of tweaks to
the kind of structure of the model um
that we're using here you could um do
and
all you'd really have to do probably is
just assign the right precisions right
to the auditory and visual input where
those are jointly
um generated by the same um
hidden cause or hidden state in the
model
um and so basically it's either you're
just inferring one cause
two causes and that either generates
receiving one flash or two flashes
um i mean anyway it would be it would be
simple to do
um and uh i guess that i don't know if i
don't
not aware of that having been done in
the kind of formal active inference mdp
literature but um
you know i mean there are known right
like neural correlates of this stuff and
you know they're in part where you'd
expect them to be right they're at like
the borders
right of like auditory and visual
processing and like association cortex
um like posterior association cortex um
and uh you know it'd be really really
really cool project right to try to

build a model of something like that and
then
just see right whether the predicted
uh neural time courses um actually
better
kind of single out right the the areas
uh the association areas where
the kind of borders division it seems
like it's
it's almost like a situated synesthesia
where you have the gabor square where
you have the frequency and time variance
but it's being one of the variants is
being hijacked by another perception
in the visual field right you know so
the
your your it that you it's crossing over
to that other sense it's very
interesting thank you thanks
um i'm gonna ask a question from the
chat and then if anyone wants to give
probably a short final note but then
we'll
end around the hour so that we can have
a whole second discussion next week
the question the chat is um to the
authors what would you say the main
limitations of the model
are or where are some next things you're
building on but what are the main
limitations of the current model and
then how are you going to be building
so i mean there's lots right uh
so i'll just list them i think we list
this in the paper somewhere too
um for starters it's a discrete model
we treat the whole visual system as
basically like one discrete hidden state

to be inferred by the high level that's
obviously like not true
but it's a good enough approximation to
actually think reproduce a bunch of
findings
it would be nice to have a multi-level
model
now with that multi-level model i think
there are some empirical cases there's
this thing called the partial awareness
hypothesis
by sid cuda where they talk about having
partial
access or the global workspace having
partial
access to different parts of the
hierarchy and that giving rise to kind
of
odd experiences so you might have
something like you might have
really good access to the fact that uh
the color of ryan's shirt
but not to the fact that it's a shirt or
vice versa
um so that would be nice i'm not sure
quite how to do that in
a hidden kind of hidden markov model
situation
um but that would be nice second thing
is
um we really just model kind of report
paradigms here so where where we're
going with this is we're
trying to extend this explicitly to
model some no report paradigms and
actually casting kind of
thinking about that in a more precise
way

and so we've actually built another
model and we're kind of writing it up at
the moment um
i mean it's not it's kind of a side
project
because it's not a part of my phd thesis
but
the idea is roughly speaking to actually
have a model that
is able to explain a lot of the weird
findings in the no report literature
and also i think we have a much more
principled we now have a much more
principled
account of report that i'm really quite
excited about
um then just like i guess on a last
limitation
would just be uh and this isn't really a
limitation because ryan has got
a lot of work in emotion and
interception on this and you know that's
somewhere that his lab is like really
pushing
um but it would just be nice if we could
use these very similar architectures and
apply
them to something like intraoceptive
awareness
and i would be i would find it
personally very compelling if you had
one i think it's too demanding to have
one model
but one mod very general model structure
that generalizes
across visual consciousness to
intraceptive awareness to like auditory
awareness

to smarter sensory awareness that would
be really nice
awesome answer ryan do you want to add
anything and then we'll have any final
comments
um you know i mean i think i think chris
covered most of the
the kinds of limitations or i mean i
don't know i mean i would just say
like like yes i mean it would be it
would definitely be nice to
do something like have a so they're what
are called mixed models
um where you have kind of a discreet
model like ours on the top
but then the um the kind of posteriors
some predictions downward that
essentially set set points
for a continuous level below it um so so
then you can have a continuous state
space where
where for instance you know it's a lot
more plausible right to think that
um these sorts of inferences that are
being made in kind of like
early visual processing or on a you know
on a continuous
continuous state space rate it doesn't
have to be like motion speed 1 motion
speed
2 right it can be it can be continuous
values
and for for things like that um
but um like another thing that i you
know i mean you know perspective
interception and
emotion i mean that's something like i
said i mean we have empirical paradigms

that we've been setting up to
to test just this kind of thing in uh in
interception
um and we have a we have a paper under
review right now where we apply a
computational model to um
interreceptive uh perception and like a
gastrointestinal interception task
that's not a multi-level model at the
moment um
and that paradigm doesn't have
um probably doesn't have enough trials
to do contrast
between um conscious versus unconscious
um stimulations but we're kind of
working on that
um so that would be nice um another
thing
that um you know chris kind of mentioned
the example of like being aware that
shirt is red but not a shirt
right that kind of thing also is really
important for modeling emotional
awareness because
um a lot of times right people might
feel right some kind of bodily
sensations you know kind of like a
pit in their stomach or like fast heart
rate or something like that right they
might be conscious of that
but not infer one step above um
and be conscious of the fact that that's
associated say with like
feeling fear right like like uh people
with like panic disorder for example
right they might
feel like a strong heartbeat but they
might instead infer oh that means i'm

having a panic
attack or something like that or they
might infer oh i'm having a heart attack
right so there are different kind of
conceptualizations that you can you can
infer
from right patterns in lower level
interceptive experience
some of which are emotional and some of
which are not um
but and this is where it becomes you
know interesting is
that um you might also prime
right an emotion category right like
like i mentioned with the guilt example
right i could prime guilt so that higher
level concept representation
um is activated but also but does not
enter into awareness right so you can
kind of have multiple levels in a
hierarchy
where each level can independently or
compete
for broadcasting um you know so
you know i've been thinking about ways
to do this model wise and the the best
thing we've come up with right now just
in terms of the practical
implementations we have would be to for
instance like have an intercepted level
um and then have the emotional level
above that but have the interception
representations also get kind of like
duplicated and passed up
to the second level so then they can
also kind of equally compete
right for access to like a third level
that would be like the the workspace

level
um but anyways i mean these are all
things that are kind of in progress but
just by way of saying um this additional
limitation is
you can only be in our model you could
only be conscious and of
conscious or unconscious of one type of
processing or
love one level of abstraction um which
is
not you know the case in the which is
not the case in the
it's not the true case
thanks a lot for these awesome answers
always a great time with the two of you
on the stream
so just thank all the participants who
joined us
live and in the calendar event there's a
forum so be helpful if you want to add
feedback on the form
and then we're going to be having 18.2
next week at the same time it's going to
be the same paper
so hopefully we'll all get another
chance to reread the paper
listen to this stream because
christopher what you said was really
deep so again much appreciated to all
and we'll see everyone next week thanks
guys